use with_columns instead of filter.

2_baseman = 4.

find all plays
where same is on 1st
then next(.shift(-1)) is == "NA"
which means a
steal failed.

on 1st = 1.

Same guy is on 2.

Ball Events
Pitch - 1
Catcher catch -
Catcher throw -

ball Events	PlayerId
1 = Pitch.	pitcher = 1
2 = Ball Acquired.	Catcher = 2.
3 = Throw.	Second bm = 4.
	1st bm = 3

batter.
one play before batter.
Play before ball in play.
player = center.
ball is acquired.

Df of 2bm.
position when.
a Steal happens.

Catch 2
Throw 3.
2nd Bm = 4
Catch = 2.

Add back in
My line up position
; find shots where.
like ~~stats~~ Steals.
are succesful US.
Fails for training.

add to my DF the next play
from my Sisters_Df

- need on batter, on_1b, on_2b.
- use lineup to get on_1b, next_on_1b
  or position? on_2b, next_on2b.

on_1st not equal to N/A.
which means someone is on 1st.

on.

Model to Predict Stolen base based on Player

1st Add batter filter for same batter
Someone on 1b but not on 2b.
Next play someone on 2b but not on 1b.

2nd

~~Filter~~ add in linevers.
Then filter by relevant
game_string : Play-per_game.
Also add 2bm Position.

filter. Someone on 1b. no one on 2b
Next play Someone on 2b not on 2b

player movement.

$\sqrt{(y_2-y_1)^2+(x_2-x_1)^2}$

- player Position Time Stamp.

- 1st thing combine Successful/Failed df 1.0/True for success 1.0 (False) for Failure
  - Then cal DCW
- 2nd add catcher then calculate catcher throw time to 2BM

ball events where catcher throws Catches then to 2bm.

(6) 2BM = 85
(6) SS = 88
(8) CF = 0
(9) RF = 0
(7) LF = 0
(3) 1BM = 52
(5) 3BM = 25.

ball_name = 73.

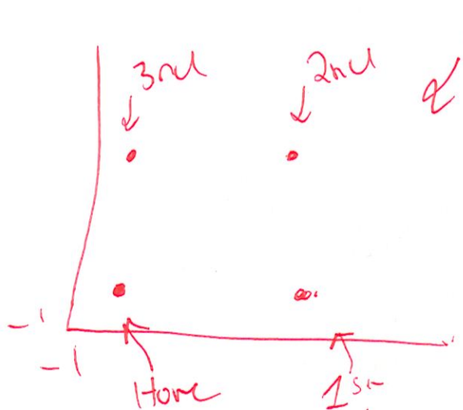
ball events. where pitcher pitch but catcher does not catch.

adding 2BM/SS. pos. to catcher time dataset agg mean for x ; y.
Then join these dataset & add to catcher stats df.

Structure.

gs	PTg	Throw time	Bmx	Bmy	Steal yes	Steal no	SS	2Brk
x-y-1	3	2.0	20	24	1	0	1	0
x-y-2	40	—	20	24	0	1	0	0
x-y-3	36	2.5	20	24	1	0	0	0
x-y-4	80	1.5	20	24	0	1	0	0

add in previous x/y



drop all Columns.
Steal=0
Throw time is null

To do
Set up ML
feature selection
for Decision tree
Drop